

HNSS Game Map: Which Games Test Which LOs

Numbers show how many items each game has on that LO: **boxes** in the Builders, **questions** in the boss battles, **creatures** in the Mon games (pathology has no Mon game). — = not covered in that game.

Coverage notes. Every LO is tested by at least two games. Only gaps: Embryomon has no creatures for **03.1** (arch timing) or **04.3** (throat & tongue innervation) — use the Embryo Builder, Pharyngeal Gauntlet and Fusion Wars. Thin spots: **07.4** (networks; lean on the Cerebral Siege's Network Oracle boss), **17.2** (Circle Siege has the most), and **12.1, 12.6, 04.10**. Each boss game's final boss mixes every LO in its lecture, and the Mon games' final stage pulls Step 1 vignettes from the boss banks. **Pathology:** every LO has builder boxes and boss questions; thinnest are **11.1** (eye anatomy review), **11.7** (optic atrophy) and **16.4** (two SCC types), which the DDx charts in the study guide and Anki deck back up.

Embryology 12 exam questions

Games: Embryo Builder · Pharyngeal Gauntlet (HNSS03) · Fusion Wars (HNSS04) · Sensory Citadel (HNSS18) · Embryomon

LO	Topic	Embryo Builder <i>boxes</i>	Boss battle <i>questions</i>	Embryomon <i>creatures</i>
HNSS03 · Pharyngeal Arches — boss battle: Pharyngeal Gauntlet				
03.1	Locate within a week the formation of the pharyngeal arches	7	5	—
03.2	Cellular and anatomical components of the arches	5	5	1
03.3	Muscular, skeletal and neural derivatives of each arch	19	6	8
03.4	Derivatives of each pharyngeal pouch	11	6	6
03.5	Derivatives of each pharyngeal cleft	4	5	2
03.6	Hox proteins in arch development	12	6	3
03.7	Alcohol vs retinoic acid: mechanisms	6	5	3
03.8	Birth defects of arch formation	7	5	4
HNSS04 · Tongue, Thyroid & Face — boss battle: Fusion Wars				
04.1	Anatomical structures of the tongue	6	5	4
04.2	Tongue development vs lung development	3	4	1
04.3	Cranial innervation of throat and tongue	7	5	—
04.4	Steps of thyroid development	5	5	3
04.5	Function of the fetal thyroid	4	4	2
04.6	Formation of the face (lips, nose, cheeks)	9	5	6
04.7	Formation of the palates	6	5	4
04.8	Neurocranium and viscerocranium	5	5	3
04.9	Clinical malformations (cleft lip and others)	10	5	4
04.10	Craniofacial ultrasound	3	3	1
HNSS18 · Eye & Ear Development — boss battle: Sensory Citadel				
18.1	Embryologic origins of eye and ear structures	16	11	12
18.2	Developmental steps: eye, inner, middle, external ear	17	10	12
18.3	Developmental abnormalities of eye and ear	9	6	5
18.X	Extra: ear structure, histology and pathology	12	6	4

Histology 8 exam questions

Games: Histo Builder · Tunic Siege (HNSS09) · Enamel Gauntlet (HNSS12) · Histomon

LO	Topic	Histo Builder boxes	Boss battle questions	Histomon creatures
HNSS09 · Eye & Ear Histology — boss battle: Tunic Siege				
09.1	Three layers of the eye: histology and function	22	17	28
09.2	Other structures of the eye (lens, lids, conjunctiva, lacrimal)	7	6	11
09.3	The three parts of the ear	7	6	12
09.4	Sensory components of the inner ear	7	6	17
HNSS12 · Oral Cavity Histology — boss battle: Enamel Gauntlet				
12.1	Structures of the oral cavity	3	4	1
12.2	Features and functions of oral structures	5	5	6
12.3	Development and organization of the teeth	9	7	19
12.4	Tongue, lingual papillae and taste buds	7	6	9
12.5	Types of salivary glands	7	6	10
12.6	Structure and function of the pharynx	3	4	2

Neuroscience 12 exam questions

Games: Neuro Builder · Cerebral Siege (HNSS07) · Neural Tube Trials (HNSS08) · Circle Siege (HNSS17) · Cerebromon

LO	Topic	Neuro Builder boxes	Boss battle questions	Cerebromon creatures
HNSS07 · The Brain — boss battle: Cerebral Siege				
07.1	Organization and basic anatomy of the brain	6	5	8
07.2	Anatomy and functions of each brain region	14	12	26
07.3	Brain disorders presented in lecture	5	5	8
07.4	Principles of brain dysfunction (networks)	1	4	2
HNSS08 · Brain Development — boss battle: Neural Tube Trials				
08.1	Stages of brain development and their failures	10	12	18
08.2	Environmental cues: teratogens and genetic malfunctions	8	10	14
08.3	Outcomes of deficits at different stages	5	5	6
HNSS17 · Vascular & Ventricular — boss battle: Circle Siege				
17.1	Blood–brain barrier and CNS entry	4	6	4
17.2	Delivery of nutrients and oxygen to the CNS	2	5	2
17.3	Major CNS arteries and what they perfuse	7	7	14
17.4	Types of CNS vascular compromise	5	6	7
17.5	Major veins and sinuses of the CNS	4	5	7
17.6	CSF production, composition, flow and clearance	5	5	7
17.7	Hydrocephalus: types, symptoms and treatment	4	5	4

Pathology 7 exam questions

Games: Pathology Builder · Fundus Fortress (HNSS11) · Aerodigestive Gauntlet (HNSS15) · Slide Sanctum (HNSS16)

LO	Topic	Pathology Builder boxes	Boss battle questions
HNSS11 · Eye Pathology — boss battle: Fundus Fortress			
11.1	Normal anatomy and histology of the eye	4	1
11.2	DDx: retinal hemorrhage, optic neuropathy, intraocular mass, proptosis	3	3
11.3	Diabetic eye disease	5	6
11.4	Retinal neovascularization	3	4
11.5	Retinal detachment	4	3
11.6	Papilledema vs optic nerve swelling	3	3
11.7	Causes of optic nerve atrophy	3	2
11.8	Primary melanoma of the eye	5	4
11.9	Retinoblastoma	6	5
11.10	DDx of orbital disease	6	6
HNSS15 · Head & Neck Pathology — boss battle: Aerodigestive Gauntlet			
15.1	Leukoplakia and erythroplakia	5	5
15.2	Causes and pathology of head and neck SCC	4	5
15.3	Clinical features and outcome of SCC	3	5
15.4	Nosebleeds and nasal obstruction	6	6
15.5	Hoarseness: laryngeal papilloma and SCC	4	5
15.6	Parotid tumors	6	6
HNSS16 · H&N Path Lab — boss battle: Slide Sanctum			
16.1	Histology of the oral cavity	2	4
16.2	Leukoplakia vs erythroplakia on the slide	2	4
16.3	SCC pathophysiology and prognosis	3	3
16.4	The two main types of SCC	2	2
16.5	Angiofibroma, papillomas, NPC, laryngeal SCC	5	5
16.6	Epistaxis vs obstruction vs hoarseness	5	5
16.7	Salivary gland tumors on the slide	6	6
16.8	Branchial cleft vs thyroglossal duct cyst	4	5